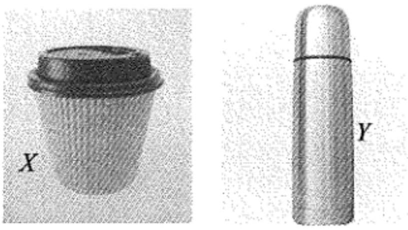
ch1 Temperature and Heat Transfer

2014 Q1

1.



Two identical scoops of ice-cream are transferred from a refrigerator into paper cup X and vacuum flask Y shown above. Under room temperature, the time required for the ice-cream in the containers to melt completely is t_X and t_Y respectively. What is the expected result and explanation ?

A. $t_X > t_Y$ as the vacuum flask reduces heat loss to the surroundings.
B. $t_X > t_Y$ as the vacuum flask retains the heat.
C. $t_Y > t_X$ as the vacuum flask keeps things cold by releasing heat into the surroundings.
D. $t_Y > t_X$ as the vacuum flask reduces the rate of heat gain from the surroundings.

2015 Q1

1. A driver turns off the engine after parking his car outdoors under the sun. Two hours later when getting into the car, he feels that the inside of the car is far hotter than outside. The best explanation is

A. the car's engine is still generating heat after the engine has been switched off.
B. the car's metal parts absorb infra-red radiation at a faster rate than the surroundings.
C. the glass windows of the car trap infra-red radiation and a greenhouse effect results.
D. the surrounding air is a good insulator of heat which reduces heat loss by conduction.

2015 Q3

3. When an object P is in contact with another object Q, heat flows from P to Q. P must have a higher

(1) temperature.
(2) internal energy.
(3) specific heat capacity.

A. (1) only
B. (3) only
C. (1) and (2) only
D. (1) and (3) only

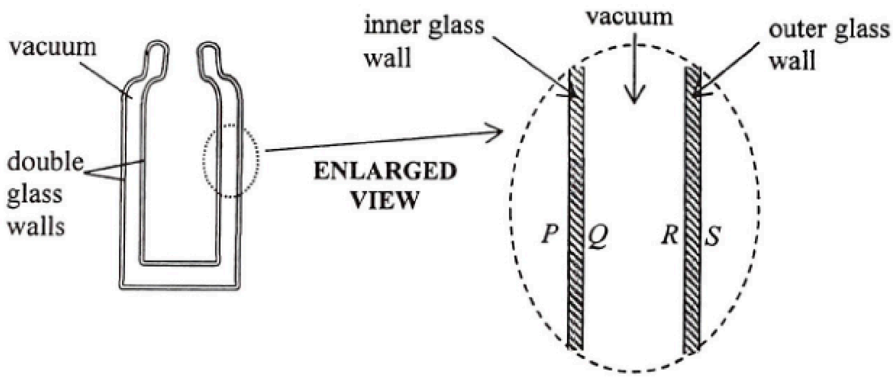
2016 Q1

1. Some icy cold liquid is kept cold inside a vacuum flask. Which statements are correct ?

(1) The flask's cork stopper reduces heat gain from the surroundings.
(2) The silver coating on the inner surface of the glass wall is a good reflector of infra-red.
(3) The vacuum between the double glass walls reduces heat gain by radiation.

A. (1) and (2) only
B. (1) and (3) only
C. (2) and (3) only
D. (1), (2) and (3)

1.

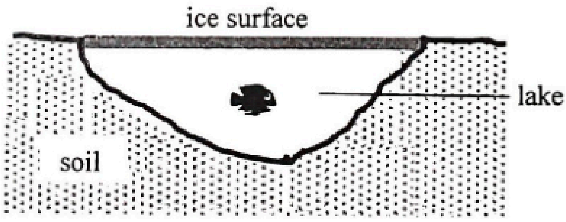


The figure shows a vacuum flask with double glass walls for keeping liquids cold. *P*, *Q* and *R*, *S* are the glass surfaces of the inner and outer glass walls respectively. Which two surfaces are usually coated with silver ?

A. *P* and *R*
B. *Q* and *R*
C. *P* and *S*
D. *R* and *S*

2018 Q2

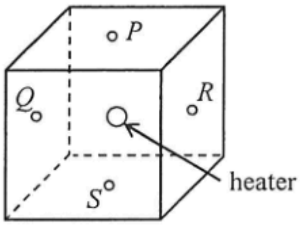
2. In some countries, the outdoor temperature can drop below 0 °C in winter such that the lake surface becomes a thick layer of ice. However the water beneath the ice surface does not easily turn into ice and most aquatic life can survive during winter time.



Which statement below best explains this phenomenon ?

A. The layer of ice provides good heat insulation.
B. The freezing point of water below the ice surface is much lower than 0 °C.
C. There is thermal energy transferred from the soil to the water in the lake.
D. Ice releases latent heat when it melts.

1. A heater is installed at the centre of a fully filled cubic water tank. Temperature sensors P, Q, R and S are fixed at the respective centres of the top, left, right and bottom surfaces of the tank.



After the heater is switched on for **a short duration**, which pair of sensors below would indicate the largest temperature difference ?

A. Q and R
B. R and S
C. Q and S
D. P and R

2. Metal blocks X and Y are identical in size and shape. X is of a higher temperature than Y. Which of the following statements must be correct ?

(1) Energy will flow from X to Y if they are in thermal contact.
(2) X is a better conductor of heat compared to Y.
(3) The total internal energy of X is higher than that of Y.

A. (1) only
B. (3) only
C. (1) and (2) only
D. (2) and (3) only

1. Which of the following statements involving heat transfer via radiation is/are correct ?

(1) Objects hotter than the surroundings do not absorb heat by radiation.
(2) The silvery surfaces of survival blankets help the body absorb heat from the surroundings by radiation.
(3) The cooling fins of a car engine should preferably be black in colour.

A. (1) only
B. (3) only
C. (1) and (2) only
D. (2) and (3) only

1. Flasks X and Y are identical except that for their outer surfaces: X is black and Y is silvery. Each flask contains the same amount of water initially at 25 °C, sealed and equipped with a temperature sensor. The flasks are then placed in the same outdoor environment at 30 °C with sunlight. After 20 minutes,

A. the water in X is hotter as a black surface is a better absorber of radiation.
B. the water in X is hotter as a black surface is a better emitter of radiation.
C. the water in Y is hotter as a silvery surface is a poorer absorber of radiation.
D. the water in Y is hotter as a silvery surface is a poorer emitter of radiation.

2026 Q2

2. The resistance of a resistance thermometer is 100.0 Ω in pure melting ice. Its resistance increases linearly with temperature by 0.385 Ω °C^{-1}. Find the temperature when the resistance is 117.2 Ω.

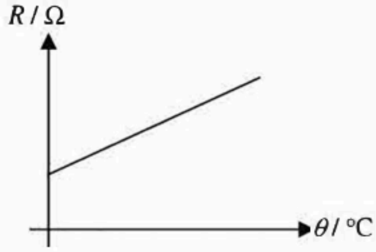
A. 6.62 °C
B. 11.5 °C
C. 44.7 °C
D. 45.1 °C

pp Q1

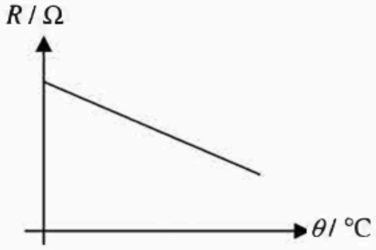
1. The graphs below show how the electrical resistances R of three different circuit elements change with temperature θ. Which of the circuit elements can be used to measure temperature ?

(1)

(2)



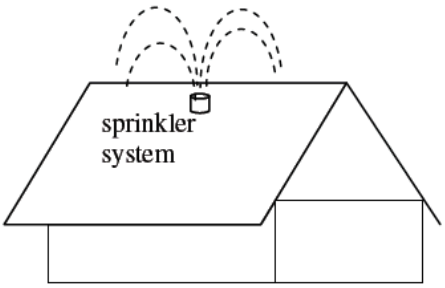
(3)



A. (1) only
B. (2) only
C. (1) and (3) only
D. (2) and (3) only

pp Q4

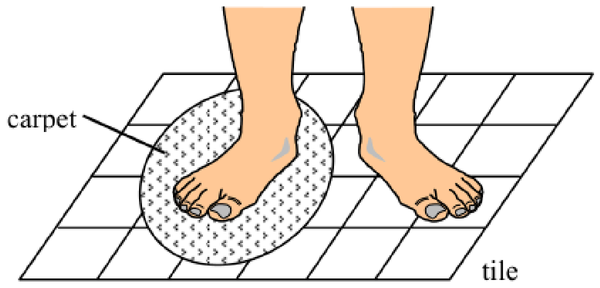
4. The sprinkler system on a rooftop is able to spray small water droplets onto the rooftop which can lower the temperature of the rooftop on hot sunny days. Which of the following explanations about the sprinkler system is/are reasonable ?



(1) Water is a good conductor, which conducts heat quickly.
(2) Water has a high specific heat capacity, absorbing a lot of energy when its temperature rises.
(3) Water has a high specific latent heat of vaporization, absorbing a lot of energy when it evaporates.

A. (1) only
B. (2) only
C. (1) and (3) only
D. (2) and (3) only

1.



Cynthia places a piece of carpet on a tiled floor. After a while, she stands in bare feet with one foot on the tiled floor and the other on the carpet as shown above. She feels that the tiled floor is colder than the carpet. Which of the following best explains this phenomenon ?

A. The tile is a better insulator of heat than the carpet.
B. The tile is at a lower temperature than the carpet.
C. The specific heat capacity of the tile is smaller than that of the carpet.
D. Energy transfers from Cynthia's foot to the tile at a greater rate than that to the carpet.